MongoDB (NoSQL) vs SQL Databases

Introduction

Databases are systems used to store, manage, and retrieve data efficiently. SQL databases are relational and structured, while NoSQL databases focus on flexibility and scalability. This presentation compares MongoDB (NoSQL) with SQL databases.

SQL Databases – Overview & Functionality

SQL databases store data in tables with rows and columns. They require a fixed schema and use Structured Query Language (SQL) for querying. They provide strong consistency and support complex relationships using joins.

Use Cases: Banking systems, school management systems, enterprise and financial applications.

MongoDB (NoSQL) – Overview & Functionality

MongoDB is a document-based NoSQL database that stores data in flexible JSON-like documents. It allows schema changes easily and is optimized for scalability and high performance in distributed systems.

Use Cases: Real-time applications, big data, content management systems, IoT, and mobile apps.

MongoDB vs SQL – Comparison

SQL databases use tables and fixed schemas, while MongoDB uses documents with flexible schemas. SQL databases scale vertically and offer strong consistency, whereas MongoDB scales horizontally and provides high availability and performance.

Conclusion

SQL databases are ideal for applications with structured data and strong consistency requirements. MongoDB is better suited for applications needing scalability and flexible data models. The choice depends on application requirements.